

BT101 - Ecology, Biodiversity & Evolution-I

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Difference between Gpp and Npp?

Gross primary productivity (GPP): Gross primary productivity is the Raw rate at which the primary producers synthesize new organic matter

Net primary productivity(NPP):

- ❖ Net primary productivity = $GPP - \text{Respiration}$
- ❖ NPP varies with seasonal changes
- ❖ Measured in mass/unit area/year

Stages of Water cycle:

1= Evaporation 2. Transpiration 3. Condensation 4. Precipitation 5. Surface Run off 6. Infiltration

Evidence of the natural selection?

Darwin based his theory on many solid evidences from his observations and collected samples.

Three major evidences:

- ❖ Geology
- ❖ Fossil Evidences
- ❖ Galápagos islands

waste water treatment ?

Wastewater treatment is a process used to remove contaminants from wastewater and convert it into an effluent that can be returned to the water cycle. Once returned to the water cycle, the effluent creates an acceptable impact on the environment or is reused for various purposes (called water reclamation).[1] The treatment process takes place in a wastewater treatment plant. There are several kinds of wastewater which are treated at the appropriate type of wastewater treatment plant. For domestic wastewater (also called municipal wastewater or sewage), the treatment plant is called a sewage treatment plant. For industrial wastewater, treatment either takes place in a separate industrial wastewater treatment plant, or in a sewage treatment plant

(usually after some form of pre-treatment). Further types of wastewater treatment plants include agricultural wastewater treatment plants and leachate treatment plants.

Trophic level process energy?

“The amount of the heat energy at each trophic level decreases as it moves through an ecosystem. As little as 10% of the energy is transferred to the next trophic level the rest is lost largely through metabolic process as a heat”.

How pollution is classified?

Pollution Classified as: 1= Point source 2= Non point source

1= Point source Single identifiable source of pollution Example: Land fill, Factory drain, Sewages pipes.

2= Non Point source Resulting from many diffuse sources Example: Sediment, Nutrients, Toxic contaminants and chemicals, Pathogens.

What are the K-Selected species and r-selected species?

K-Selected species: K-selected species possess relatively stable populations fluctuating near the carrying capacity of the environment. These species are characterized by having only a few offspring but investing high amounts of parental care. Elephants, humans, and bison are all k-selected species.

r-selected species:

r-selected species are those that emphasize high growth rates, typically exploit less-crowded ecological niches, and produce many offspring, each of which has a relatively low probability of surviving to adulthood. Examples: bacteria, grasses, cephalopods, rodents.

What is the difference between Analogy & homology?

Homology:

Homology involves the study of organs that have evolved from same origin or from common ancestor but have different functions. It is a Divergent evolution.

For example:

- ❖ Wings of bat and Hands of humans.

Analogy:

Analogy involves the study of organs that are evolved from different ancestors but perform a similar function. It is a Convergent Evolution.

For example:

- ❖ Wings of bat and wings of insects.

Characteristics of thermosphere:

It is located above the mesosphere and is the fourth layer of the earth. Its temperature changes with solar activity. In its interior sublayer i.e. ionosphere the radio waves are reflected. The boundary between the thermosphere and the exosphere is called Thermopause. The thermosphere is very high above the Earth and is distinguished by an increase in temperature with altitude. The density of the air is extremely low, but the activity of the molecules are very high due to the amount of energy they are receiving from the sun. Individual molecules can be as warm as 2000 degrees C.

Differentiate between top-down and bottom-up effect.

1= Top-Down Effect: Effect flows down through a trophic chain. From apex predator to lower trophic levels

2= Bottom-Up Effect: Effect flows up through a trophic chain. From primary producers to higher trophic levels.

Define transpiration, ground water and percolation?

Transpiration: Plants absorb water from the soil. Water moves from the roots through the stems to the leaves. The process of evaporation through plant leaves is called transpiration.

Groundwater: is the water found underground in the cracks and spaces in soil, sand and rock. It is stored in and moves slowly through geologic formations of soil, sand and rocks called aquifers.

Percolation (Infiltration): Percolation is an important process. Rain water soaks into (infiltrates) the ground, into the soil and underlying rock layers. Some of this water ultimately returns to the surface at springs or in low spots downhill. Some of the water percolates underground. It is called groundwater.

Flow of Energy in Ecosystem Feeding Levels: or Explain Feeding level of organisms:

- ❖ Food energy passes through different sequence.
- ❖ Example: plants → grass hopper → frog → snake → hawk
- ❖ Many plant species eaten by insects
- ❖ Many insects species eaten by frogs
- ❖ this complexity is organized by ecologists.
- ❖ Recognized limited number of feeding or trophic levels
- ❖ In Food Chain
- ❖ Food energy passes through different sequence.
- ❖ Level 1: Primary producer
- ❖ Level 2: Primary consumer
- ❖ Level 3: Secondary consumer
- ❖ Level 4: Tertiary Consumer Level
- ❖ 5: Apex Predator

Ozone depletion:

Ozone depletion is the destruction of ozone in the stratosphere, where it shields the earth from harmful ultraviolet radiation. The substances that are responsible for ozone depletion are Chlorofluorocarbons, Hydro chlorofluorocarbons, Solvents, Halocarbon refrigerants, Methyl bromide, Carbon tetrachloride, Methyl chloroform.

Energy flow process?

Small fraction of solar energy (1%) is consumed in photosynthesis. Transformed into chemical bond energy. 99% solar energy is converted into heat. Chemical-bond energy is passed from one heterotroph trophic level to the next. Loss of 90% energy from one level to next.

Theory of law of inheritance:

Gregor Mendel, through his work on pea plants, discovered the fundamental laws of inheritance. He deduced that genes come in pairs and are inherited as distinct units, one from each parent. Mendel tracked the segregation of parental genes and their appearance in the offspring as dominant or recessive traits.

Write a note on carbon:

Carbon is the main component of living organisms. 20% of human body is carbon by weight. CO₂ is the major source of carbon. Atmosphere contains about 750 billion metric tons of carbon in form of CO₂. Photosynthetic organisms take in CO₂ from the atmosphere and synthesize the carbon-containing organic compounds. This process is sometimes called carbon fixation. Animals eat plants and make their own organic compounds. Production of energy by breakdown of these organic compounds in respiration.

Secondary productivity:

The rate that new organic matter is made by means of individual growth and reproduction in all the herbivores. Each heterotroph trophic level has its own secondary productivity.

Theory of natural selection:

The theory of evolution by natural selection, first formulated in Darwin's book "On the Origin of Species" in 1859, is the process by which organisms change over time as a result of changes in heritable physical or behavioral traits.

What are the Autotrophs:

An autotroph is an organism that can produce its own food using light, water, carbon dioxide, or other chemicals. Because autotrophs produce their own food, they are sometimes called producers. Plants are the most familiar type of autotroph, but there are many different kinds of autotrophic organisms.

Types: 1= Photoautotrophs, 2= Chemoautotrophs

1= Photoautotrophs: a photosynthetic organism (such as a green plant or a cyanobacterium) that utilizes energy from light to synthesize organic molecules. Examples: plants, algae, and cyanobacteria

2= Chemoautotrophs: Chemoautotrophs occur in dark habitats. They obtain energy by means of inorganic oxidation reactions. Convert toxic molecules into glucose Example: Microbes, mussels

What are the aposematism and Invasive species?

Animals that warn predators of their dangerous nature are called **aposematic**. So the advertising by an animal to potential predators that it is not worth attacking or eating. Lionfish advertise their venomous spines with waving flags and banners. Bright or contrasting color patterns, such as the yellow and black stripes of a wasp, serve as common aposematic signals.

Invasive species:

Colonization, a natural process by which a species expands its geographic range, occurs in many ways. Examples: Seed dispersal by birds, Lowering of sea levels join two isolated land masses, Flooding.

Types of invasive species: 1= Animals 2= Plants 3= Pathogens: They cause an imbalance in the natural environment. They outcompete the native species.

Difference between Idiobiont and koinobiont:

Idiobiont: Idiobiont parasitoids are species that attack hosts that do not feed or grow, such as eggs or pupae, or hosts which are permanently paralyzed by the female parasitoid before oviposition

Koinobiont parasitoids:

Koinobiont parasitoids attack hosts that continue feeding and growing during parasitism, whereas idiobiont parasitoids attack non-growing host stages or paralyzed hosts.

Extinction and its causes:

Termination of an organism or of a group of organisms (taxon), usually a species. Moment of extinction is death of the last individual of the species. Capacity to breed and recover may have been lost before this point.

Variety of causes, independently or in concert, are responsible for extinctions. • Disruption of habitat • Loss of ecosystem interactions e.g. disrupted food chain or web, symbiosis, predation • Pollution • Loss of genetic variation • Catastrophic disturbances • Natural or human-caused activities Habitat loss is the most important cause of modern-day extinction. Human activities are negatively affecting all types of habitat, from rain forest to ocean floor

Natural habitats may be adversely affected by humans in three ways.

1. Habitat destruction
2. Pollution
3. Habitat fragmentation

Ozone Depleting Substances:

Halocarbon refrigerants , Solvents , Propellants in aerosols, Chlorofluorocarbons, Hydro chlorofluorocarbons, Methyl bromide, Carbon tetrachloride, Methyl chloroform.

Logistic growth: 5 marks

A logistic growth curve is an S-shaped (sigmoid) curve that can be used to model functions that increase gradually at first, more rapidly in the middle growth period, and slowly at the end, leveling off at a maximum value after some period of time. If we look at a graph of a population undergoing logistic population growth, it will have a characteristic S-shaped curve. The population grows in size slowly when there are only a few individuals. Then the population grows faster when there are more individuals. Finally, having lots of individuals in the population causes growth to slow because resources are limited. In logistic growth, a population will continue to grow until it reaches carrying capacity.

Write the sources of waste water:

Human excreta (feces, urine, blood and other bodily fluids) often mixed with used toilet paper or wet wipes; this is known as blackwater if it is collected from flush toilets Washing water (personal hygiene, clothes, floors, dishes, cars, etc.), also known as greywater or sullage Surplus manufactured liquids from domestic sources (drinks, cooking oil, pesticides, lubricating oil, paint, cleaning detergents, etc.) 1. Domestic wastewater from households 2. Municipal wastewater from communities (also called sewage) 3. Industrial wastewater from industrial activities 4. Agricultural wastewater from agricultural activities 5. Storm water.

Layers of atmosphere:

- ❖ Troposphere: 0 to 12 km
- ❖ Stratosphere: 12 to 50 km
- ❖ Mesosphere: 50 to 80 km
- ❖ Thermosphere: 80 to 700 km
- ❖ Exosphere: 700 to 10,000 km

Global warming effects:

- ❖ More frequent and severe weather. Higher temperatures are worsening many types of disasters, including storms, heat waves, floods, and droughts.
- ❖ Higher death rates.
- ❖ Dirtier air.
- ❖ Higher wildlife extinction rates.
- ❖ More acidic oceans.
- ❖ Higher sea levels.

Conventional current:

The conventional current can be defined as charge per unit time transported in a certain direction. In conventional current, the flow of electrons is assumed as a flow of protons in the opposite direction.

Write a note on acid rain:

Acid rain is caused by a chemical reaction that begins when compounds like sulfur dioxide and nitrogen oxides are released into the air. These substances can rise very high into the atmosphere, where they mix and react with water, oxygen, and other chemicals to form more acidic pollutants, known as acid rain. Sulfur dioxide and nitrogen oxides dissolve very easily in water and can be carried very far by the wind. As a result, the two compounds can travel long distances where they become part of the rain, sleet, snow, and fog that we experience on certain days.

Human activities are the main cause of acid rain. Over the past few decades, humans have released so many different chemicals into the air that they have changed the mix of gases in the atmosphere. Power plants release the majority of sulfur dioxide and much of the nitrogen oxides when they burn fossil fuels, such as coal, to produce electricity. In addition, the exhaust from cars, trucks, and buses releases nitrogen oxides and sulfur dioxide into the air. These pollutants cause acid rain.

Benefits of atmosphere?

The Earth's atmosphere protects and sustains the planet's inhabitants by providing warmth and absorbing harmful solar rays. In addition to containing the oxygen and carbon dioxide, which living things need to survive, the atmosphere traps the sun's energy and wards off many of the dangers of space.

Limiting factors regulate population size:

- ❖ Rate of Birth
- ❖ Rate of Death/mortality
- ❖ Dispersal
- ❖ Food Resources
- ❖ Space to grow
- ❖ Environmental factors

Water pollution:

Any physical or chemical change in water that adversely affects the health of humans and other organisms.

Categories:

- ❖ Sewage
- ❖ Disease-causing agents
- ❖ Sediment pollution
- ❖ Inorganic plant and algal nutrients
- ❖ Organic compounds
- ❖ Inorganic chemicals
- ❖ Radioactive substances
- ❖ Thermal pollution

Causes water pollution:

The main water pollutants include bacteria, viruses, parasites, fertilisers, pesticides, pharmaceutical products, nitrates, phosphates, plastics, faecal waste and even radioactive substances. These substances do not always change the colour of the water, meaning that they are often invisible pollutants.

Difference between grazing and browsing ?

1= Grazers Feeds on plants such as grasses, soft flowering plants (forbs) and algae. Plants are not killed. Examples: Deer & Snails

Types of Grazing:

1. **Graminivory** Feeding primarily on grass. Specifically "true" grasses in family Poaceae. Examples: Horses, cattle.
2. **Coprophagy** Feeding on grass, forbs, leafy weeds, fruits, tree bark Heavy grazing for half an hour of grazing period Followed by half an hour of selective feeding. Example: Rabbit

2= Browsers Feed on non-grass plant material Leaves, soft shoots, twigs, fruits, woody plants such as shrubs Examples: Goats, deer, giraffe, elephants.

How many form of energy in ecosystem:

Energy exists in several different forms 1. Light 2. Chemical-bond energy 3. Motion 4. Heat

Define Aestivation:

Aestivation is also called “summer sleep”, is the Survival in extended period of drying. It is the low metabolic process by the animals during summers. Examples: Insects, frog, turtle

Carbon cycle:

- ❖ Carbon enters the atmosphere as CO₂.
- ❖ CO₂ is absorbed by autotrophs such as green plants.
- ❖ Animals consume plants, thereby, incorporating carbon into their system.
- ❖ Animals and plants die, their bodies decompose and carbon is reabsorbed back into the atmosphere.

Food chain:

A food chain is a linear network of links in a food web starting from producer organisms and ending at an apex predator species, detritivores, or decomposer species. A food chain also shows how organisms are related to each other by the food they eat. Each level of a food chain represents a different trophic level.

Primary and secondary treatment of water ?

Primary Treatment: it is a Mechanical process that include:

- ❖ Screening
- ❖ Grinding
- ❖ Grit Chamber (heavy particles - “grit”)
- ❖ Sedimentation Tank (suspended solids - “sludge”)
- ❖ Chlorination of effluent

Secondary Treatment: it is a Biological process that Uses bacteria to remove organic matter dissolved in water (ODW) and lower BOD.

Types of parasites:

1. Microparasites: reproduce inside host. Examples: Bacteria, viruses
2. Macroparasites: release juvenile outside host. E.g. trematodes
3. Endo parasites: Examples: Liver fluke, tape worm, nematodes
4. Ectoparasites: Examples: Ticks, mites, fleas, mosquito, lice

what are the types of the consumers:

primary consumers, secondary consumers, tertiary consumers or apex consumers are the different types of consumers. These types are according to the trophic level to which they belong.

Note on decomposition:

Decomposition is the process by which dead organic substances are broken down into simpler organic or inorganic matter such as carbon dioxide, water, simple sugars and mineral salts. The process is a part of the nutrient cycle and is essential for recycling the finite matter that occupies physical space in the biosphere. Bodies of living organisms begin to decompose shortly after death. Animals, such as worms, also help decompose the organic materials. Organisms that do this are known as decomposers. Although no two organisms decompose in the same way, they all undergo the same sequential stages of decomposition. The science which studies decomposition is generally referred to as taphonomy from the Greek word taphos, meaning tomb. Decomposition can also be a gradual process for organisms that have extended periods of dormancy.

What are the fragmentation species:

Fragmentation, also known as a splitting method of reproduction and is seen in many organisms such as cyanobacteria, fungi, many plants, and also in animals including flatworms, sponges, some annelid worms and sea stars.

What is autolysis and putrefaction:

Autolysis: Breaking down of tissues by the body's own internal chemicals and enzymes

Putrefaction: Breakdown of tissues by bacteria. And releases compounds such as cadaverine and putrescine causing decaying odor

Limiting factor effect the size of population:

Limiting factors are resources or other factors in the environment that can lower the population growth rate. Limiting factors include a low food supply and lack of space. Limiting factors can lower birth rates, increase death rates, or lead to emigration.

Limiting factors regulate population size:

- ❖ Rate of Birth
- ❖ Rate of Death/mortality
- ❖ Dispersal
- ❖ Food Resources
- ❖ Space to grow
- ❖ Environmental factors

Types of ecological pyramids:

Ecological Pyramid refers to a graphical (pyramidal) representation to show the number of organisms, biomass, and productivity at each trophic level.

There are three different types of ecological pyramids to illustrate ecosystems:

1. Pyramid of energy: represents how much energy is available in each trophic level
2. Pyramid of numbers: represents the actual number of organisms present in each trophic level
3. Pyramid of biomass: represents the total mass of living things in each trophic level

Characteristics of habitat:

A habitat is a place or region where an organism lives, or the area where one would go to find it. Habitats are natural environments with specific conditions that sustain animal and plant populations. This means that a habitat will have all the requirements for an organism to gather food, reproduce, and take shelter within it. In general, habitats are characterized by different biotic and abiotic factors, such as water, soil, topography, moisture, temperature and food availability. These are the fundamental factors that determine the distribution of organisms and habitats across the world.

Habitats can be divided into two groups: (i) terrestrial and (ii) aquatic. Terrestrial habitats include forests, grasslands and deserts, while aquatic habitats include freshwater habitats (marshes, rivers, lakes) and marine habitats (sea, salt marshes).

Define Light pollution and its impacts:

Light pollution, or artificial light at night, is the excessive or poor use of artificial outdoor light, and it disrupts the natural patterns of wildlife, contributes to the increase in carbon dioxide (CO₂) in the atmosphere, disrupts human sleep, and obscures the stars in the night sky.

Impacts: 1= The degradation of photic habitat by artificial light, 2= Due to sources of light, which compromises human health, 3= The introduction by humans, directly or indirectly, of artificial light into the environment.

Consumers resources system:

Consumer–resource interactions are the core motif of ecological food chains or food webs, and are an umbrella term for a variety of more specialized types of biological species interactions including prey-predator (see predation), host-parasite (see parasitism), plant-herbivore and victimexploiter systems.

Top-Down Effect: Effect flows down through a trophic chain. From apex predator to lower trophic levels Bottom-Up Effect: Effect flows up through a trophic chain. From primary producers to higher trophic levels.

Which microorganism play* *role in nitrogen fixation*

Bacteria play a central role: Nitrogen-fixing bacteria, which convert atmospheric nitrogen to nitrates. Bacteria of decay, which convert decaying nitrogen waste to ammonia. Example **cyanobacteria**

Define omnivores:

An omnivore is an organism that eats plants and animals. The term stems from the Latin words omnis, meaning “all or everything,” and vorare, meaning “to devour or eat.” Example: humans.

Define nitrogen fixation;

Nitrogen fixation is the process by which nitrogen is taken from its molecular form (N₂) in the atmosphere and converted into nitrogen compounds useful for other biochemical processes. Fixation can occur through atmospheric (lightning), industrial, or biological processes.